Jonathan Gutman

A Means-End Chain Model Based on Consumer Categorization Processes

To practitioner and researcher alike, consumer values play an important role in understanding behavior in the marketplace. This paper presents a model linking perceived product attributes to values.

VALUES have been shown to be a powerful force in governing the behavior of individuals in all aspects of their lives (Rokeach 1968, Yankelovich 1981). Their use in marketing planning and strategy could be improved if the behavior of consumers could be related to their values. Several attempts have been made to provide a theoretical and conceptual structure connecting consumers' values to their behavior (Howard 1977, Vinson, Scott and Lamont 1977, Young and Feigin 1975). These attempts can be subsumed under the rubric of a means-end chain. Means are objects (products) or activities in which people engage (running, reading). Ends are valued states of being such as happiness, security, accomplishment. A means-end chain is a model that seeks to explain how a product or service selection facilitates the achievement of desired end states. Such a model consists of elements that represent the major consumer processes that link values to behavior.

The means-end chain concept offers marketing managers a way to position products by associating means (the physical aspects of products) with advertising that seeks to tie the consumption of products to the achievement of desired ends (valued states). For example, knowing that consumers want to look well dressed doesn't tell us much unless we know why they want to look that way (sexual attractiveness, accomplishment, neatness, etc., which are value-level considerations) and what attributes in clothing they associate with being well dressed. For the means-end chain to have a central place in marketing planning and consumer research, the nature of the elements comprising means-end chains and the nature of the connections between these elements must be specified, and the model must be operationalized so as to facilitate research.

Jonathan Gutman is Associate Professor, University of Southern California.

Conceptual Model for Means-End Chain

The purpose of this paper is to present the author's means-end chain model. The model is based on two fundamental assumptions about consumer behavior: (1) that values, defined here as desirable end-states of existence, play a dominant role in guiding choice patterns, and (2) that people cope with the tremendous diversity of products that are potential satisfiers of their values by grouping them into sets or classes so as to reduce the complexity of choice. This suggests that in addition to the product-class type of product

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categories, e.g., toothpaste, mouthwash, chewing gum, mints, consumers are capable of creating categories based on product functions. For instance, they could produce a category of breath fresheners that includes products from all of these product classes ("fresh breath" is unlikely to be classed as a value per se, but it might lead to favorable interpersonal interactions that could lead more directly to valued states).

In addition to these two assumptions about consumer behavior that are essential to the particular form of the model, there are two other assumptions of a more general nature. These assumptions are that all consumer actions have consequences (although all consumers wouldn't agree that the same actions in the same situations produce the same consequences), and that consumers learn to associate particular consequences with particular actions.

Consequences

Consequences may be defined as any result (physiological or psychological) accruing directly or indirectly to the consumer (sooner or later) from his/her behavior. Consequences can be desirable or undesirable. There is a term and a literature in marketing dealing with desirable consequences called benefits (Haley 1968, Myers 1976), which are the advantages consumers enjoy from the consumption of products. Benefits differ from attributes in that people receive benefits whereas products have attributes. "White teeth" is a benefit that can be obtained from brushing your teeth with any one of a number of toothpastes or powders with "teeth whiteners."

Consequences may be physiological in nature (satisfying hunger, thirst, or other physiological needs), or the nature of the consequences resulting from one behavior may be psychological (self-esteem, improved outlook for the future) or sociological (enhanced status, group membership). Direct consequences come directly from the thing consumed or from the act of consumption. Indirect consequences can occur when other people react favorably or unfavorably to us because of our consumption behavior. They can also occur when direct consequences produce other consequences. For instance, buying a new dress makes you feel better (a direct consequence); because you feel better, people react more favorably to you (an indirect consequence). In addition, some consequences occur immediately during the act of consumption (having one's thirst quenched), whereas others occur later (indigestion, tooth decay).

The central aspect of the model is that consumers choose actions that produce desired consequences and minimize undesired consequences. It is suggested, following Rokeach (1973), that values provide consequences with positive or negative valences. Therefore, the values-consequences linkage is one of the critical linkages in the model. To the extent that values are ordered in importance (Rokeach 1973), they also give consequences importance, in that consequences leading to important values should be more important to a person than those leading to less important values (a position very similar to that espoused by Rosenberg 1956).

An act of consumption must occur in order for the desired consequences to be realized. Thus a choice among alternative products has to be made. In order to make this choice, the consumer has to learn which products have attributes that will produce these desired consequences. Therefore, the second important linkage in the model is that between consequences and product attributes.

Product-Use Situation

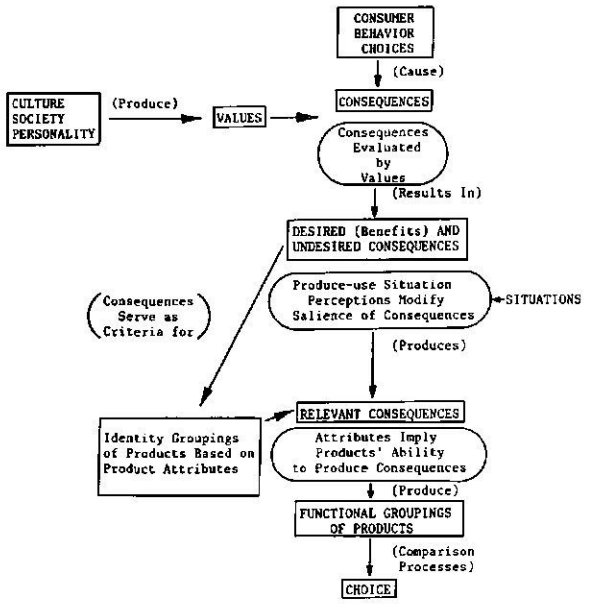
The last concept necessary to the explication of the model at this point is that of the product-use situation. Consumers obviously encounter many potential product-use situations. There are many ways of defining and describing such situational variables (Belk 1975, Fennell 1978). But for the level of analysis referred to in this context we may simply define a product-use situation as any situation that involves the use of a commercially available product or service.

Consumers have desired consequences they are trying to achieve. Consumption situations provide them with an opportunity to achieve these consequences. Each consumer learns over time which choices in a given situation produce these desirable consequences and which do not. Consumers evaluate product-use situations in terms of their potential impact over time. The length of the time horizon the consumer adopts with respect to the situation can modify the importances of the consequences. Thus someone who doesn't eat rich desserts on a regular basis for health or caloric reasons may nevertheless indulge in a chocolate mousse on a special occasion. The means-end chain permits us to focus on the basic aims consumers have in life while not losing sight of how these aims influence choices in specific situations.

Summary of Conceptual Model

To summarize, the model may be conceptualized as shown in Figure 1. Consumer values give consequences valence and importance. These importances are modified by the situation, which induces the consumer to consider the consequences in light of the demands of the situation. The relevant consequences coming out of this person-situation interaction are the basis of a functional category of products that can best produce the relevant consequences. These products are selected on the basis of attributes they possess,

FIGURE 1
Conceptual Model for Means-End Chain



which imply their ability to produce the desired consequences and avoid the undesired consequences.

Over time, consumers learn to distinguish between products they wouldn't use and those they would use and in which types of situations they would use those products. To understand the linkages between the levels in the means-end chain, the differences as well as the similarities between the products in the functional categories have to be studied. The ways in which consumers describe the diversity of these functionally equivalent products can tell us much about their means-end chains. This approach also allows us to determine why certain products are not chosen or considered and the consequences that would be produced if they were chosen.

Consider, as an example, the purchase of a hot dog. What consequences are produced by the consumption of a hot dog and how do these consequences relate to values? There are many potential consequences depending on who is doing the consuming: taste experience, type of meat, nutrition obtained from meat, preparation consideration, consumption of additives, salt, fat, and so forth.

Relevant values with respect to pleasure, living a comfortable life, religion, and good health, among others, could play a role in attaching valences and importances to these consequences. Different types of hot dogs, luncheon meats or sandwich fillings—depending on the breadth of alternatives the consumer might wish to consider—would be categorized at an initial grouping level based primarily on their physical characteristics. The usage situation (regular lunch, occasional lunch, at a baseball game, etc.) would modify the importances of the consequences, producing a set of relevant consequences for that consumption situation. In addition, this set of relevant consequences would be applied to the product class groupings that comprise the categories at the initial grouping level. At this point a final choice could be made, if more than one choice remained.

Categorization Processes

The categorization process is proposed as the method of studying how consumers organize their thinking about specific product alternatives. It is hypothesized that consumers create arrays of products that will be instrumental in helping them achieve their desired consequences, which in turn move consumers toward valued end states. If this connection can be made, marketing will be in a better position to understand how personal values influence everyday consumer choices.

Categorization processes represent the way in which consumers segment their environments into meaningful groups by creating equivalences among nonidentical stimuli. "The world consists of an infinite number of potentially different stimuli. Thus a basic task of all organisms . . . is a segmentation of the environment into classifications by means of which nonidentical stimuli can be treated as equivalent" (Rosch 1978, p. 1). Categorizing is the way consumers separate the world into smaller units to make it manageable, to limit it to the point where it matches their individual capacities.

The marketplace is full of many more objects than individuals have values. Therefore, "ends" are few, and "means" are many (Vinson, Scott and Lamont 1977, p. 46). It is essential for consumers to reduce the complexity inherent in the multitude of alternatives with which they are faced. "When faced with a set of stimuli people often organize them in clusters to reduce information load and facilitate further processing" (Tversky and Gati 1978, p. 91). To enable a person to achieve his/her values, groups or categories of products (product classes) have to be related systematically to the higher-level ends if the chain is to serve its instrumental purpose.

This occurs because although grouping is determined by the object's properties, the choice of properties to be focused on is influenced by values. Consumers group products in different categories depending on which features they emphasize and which features they ignore. In this manner, the ways in which products are identified or described by consumers fit their subsequent grouping of these products into functional groupings (for example, one consumer might have a

category of luncheon meats, another of pork products, another of meat with and without additives). This means that values are translated from their context at the more inclusive levels of the chain to the less inclusive levels where products are categorized into product classes. Consumers are more likely to be in agreement on what physical characteristics products possess than at high levels of inclusiveness where categories are based on the function of value-producing consequences.

This categorization process takes place at each level of the means-end chain as categories of greater inclusiveness are formed at higher and higher levels of the chain (inclusiveness refers to the degree of similarity among objects in a category). Brands of paper towels grouped together would represent a category of low inclusiveness. If consumers distinguish between plain paper towels and those with a pattern, two categories of lower inclusiveness would be created. If paper towels were grouped with paper napkins, cups, and tissues, a more inclusive category of paper products would be created. And depending on what other items are grouped with paper products, people might form any of a variety of functionally labeled categories, such as "saves me time and energy," "keeps me healthy," "uses up scarce resources." It is in this way that products are categorized and create groupings that are optimal in providing desired consequences that will provide value satisfaction.

It will be argued that these categories are based on a cognitive element called a distinction. Distinctions are dichotomies that represent *the end points of dimensions along which objects may be compared*. If two products can be compared using a hot/cold distinction, this indicates that they may be compared by temperature, although they may differ. An array of products can be created by applying this distinction to any of a limitless number of products. However, many products may not be comparable using this distinction. If a consumer doesn't think of a chain saw, a dry cleaner or a box of paperclips as being either hot or cold, then none of these objects would be part of the array. Other products such as orange juice, coffee, tea, milk, water, corn flakes or oatmeal could be compared on this basis.

The resulting array can contain as many categories as there are products or just one category (if all objects were thought to be the same temperature). The essential point is that *the distinction being dichotomous has nothing to do with the nature of the array of products created by its application*. A distinction is not an array of products; it is a dichotomous cognitive element in which the two parts of the dichotomy have some meaningful relation to each other. What is meaningful is a personal decision for each consumer to make.

Having presented a conceptual model for a means-end chain, other means-end approaches that have appeared in the literature will be reviewed. It will be shown that existing approaches have fallen short of meeting one or another of the criteria of specifying the links in the chain, of showing how they are connected to one another, or of being sufficiently operationalized to guide research. In most cases the central consideration in developing these models is not the presentation of a fully developed means-end chain model, either because the focus is broader (Howard 1977), narrower (Vinson, Scott and Lamont 1977), or wholly pragmatic (Young and Feigen 1975).

Review of Means-End Model Literature

Values Orientations for Means-End Chains

Rokeach (1968, 1973) has provided much of the impetus for research on values. "A value is an enduring belief that a specific mode of conduct or end-state of existence is personally or socially preferable to an opposite or converse mode of conduct or end-state of existence" (1973, p. 5). Thus values are special kinds of preferences for modes of conduct or end-states of existence. All behavior is not guided by values, only that behavior that is related to maintaining and enhancing self-esteem (Rokeach 1973, p. 14). A person's values are assumed to come from "culture, society, and its institutions, and personality" (1973, p. 3). After a value is learned it is integrated into an organized system of values in which all of a person's values are ordered in importance with respect to one another.

Two kinds of values have been defined by Rokeach: terminal and instrumental. Terminal values are concerned with preferred end-states of existence (i.e., happiness, security, accomplishment), and instrumental values are related to modes of behavior (i.e., honest, courageous, broad-minded) which are instrumental in achieving these end-states.

The interrelation between these "means" values and these "ends" values is referred to as a value system. Rokeach's interest in values has not led to his developing mechanisms for translating instrumental values into choices of objects to satisfy those values. Therefore, from a marketing perspective more needs to be done to demonstrate how preferred end-states of existence are translated into specific choices in specific situations. The author's model does not make use of instrumental values but uses consequences more akin to terminal values than instrumental values. A consequence is a state of being produced by an act of consumption. It is not an end state, but its relation or ability to move the consumer toward an end state is

what gives the consequence a meaningful role in the model.

Vinson, Scott and Lamont's Centrality of Beliefs

The nature of the various means-end chains appearing in the literature is a function of the perspectives of their authors. Vinson, Scott and Lamont (1977) have stayed closest to Rokeach in providing a model of the consumer's value-attitude system. Their model has global values (a redefinition of terminal values) at the most central level (closest to a person's self-concept). Domain-specific values, preferences for something less general than end-states of being, are at an intermediate level. And evaluative beliefs about products are at the least central level. Domain-specific values are hypothesized to bridge "the gap between the traditional conception of closely held but general global values and the less closely held descriptive and evaluative beliefs about products' attributes" (Vinson, Scott and Lamont 1977, p. 45).

Values are defined as "centrally held cognitive elements that stimulate motivation for behavioral response" (Vinson, Scott and Lamont 1977, p. 45), that "exist in an interconnected hierarchical structure in which global values are related and connected to generalized consumption-related values, which are, in turn, similarly associated with product attributes" (Vinson, Scott and Lamont 1977, p. 49). However, although various levels of a means-end chain are articulated, no means are provided by which values at global and domain-specific levels are related to each other nor how either level of values is tied to evaluations of product attributes.

Howard's Means-End Chain Model

Howard suggests that the total meaning of a brand is called its semantic structure (see Figure 2), which refers to a hierarchically organized set of categories into which a brand is placed by a consumer (Howard 1977, p. 42). The structure is hierarchical because categories at the top (viz., breakfast beverages) subsume categories at lower levels (tea, coffee, milk and juices are types of breakfast beverages and regular, instant, and freeze-dried are subsumed by coffee). As these categories are based on semantic properties (what things are called) rather than consumers' evaluations of them, there is likely to be high agreement among people as to the meaning of a brand.

However, choice is more a matter of evaluation than merely being based on semantic meaning, so Howard posits a hierarchical evaluative structure having two levels, one headed by terminal values and one headed by instrumental values (Howard 1977, pp. 94, 163):

Terminal Values	choice criteria for product class	belief about product class	attitude toward product class
Instrumental Values	choice criteria for brand	belief about brand	attitude toward brand

Terminal and instrumental values are defined in the same way that Rokeach defines them. Howard defines choice criteria as "mental counterparts of the attributes by which a consumer judges a brand" (1977, p. 28). Beliefs indicate where a product lies with respect to a given choice criterion. Attitude is conceptualized as beliefs weighted by the saliences given them by choice criteria.

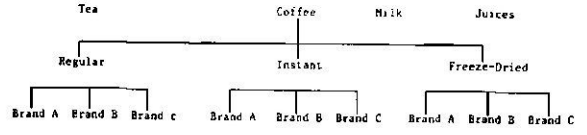
What Howard is suggesting is that grouping and distinguishing will result in semantic categories, and that choice criteria generated by terminal values enable consumers to choose appropriate product classes, while choice criteria generated by instrumental values enable consumers to pick appropriate brands.

The author's model differs from Howard's model in that the concept of instrumental values is not used. Values, as preferred end-states of being, are a type of consequence for which a person has no further (higher) reason for preference. Values and consequences are bases for categorization along with initial grouping categories based on physical characteristics (Howard's semantic meaning). Another key difference is the author's reliance on categorization as the basis for levels in the means-end chain. There is only one chain as opposed to hypothesizing parallel semantic and evaluative chains. Value and consequence level categories subsume grouping level categories, creating a single hierarchy of categories across all levels of the means-end chain.

The Grey Benefit Chain

Young and Feigin (1975) discuss benefit chain analysis as a method for linking emotional or psychological benefits to product claims or product attributes. A benefit chain begins with a product description including a specific product attribute. The consumer provides two benefits that are derived from this at-

FIGURE 2
Hypothetical Semantic Structure for Coffee as a Breakfast Beverage



tribute and two benefits that are derived from each of the two initial benefits. The process is repeated once more, yielding 14 benefits in all. The links in the chain can be demonstrated by presenting one of their examples (Young and Feigin 1975, p. 73):

The Product	Functional Benefit	Practical Benefit	Emotional Pay-Off
hair spray that holds and leaves hair soft	leaves hair easier to manage	I don't need to spend so much time on hair	leaves me free to do other things I want to do

This approach directly attacks the problem of explicating the means-end chain (the author would define functional and practical benefits and emotional payoffs as types of consequences). It seeks associations between surface properties of products and benefits that are successively closer to the core values of the consumer.

The Grey Benefit Chain, although it focuses only on obtaining verbalizations associated with various levels in the chain, offers the best view of the power of the means-end approach to provide a structure for solving marketing problems at several levels simultaneously. However, this approach to means-end chain development needs an infusion of theory to give it a broader perspective in linking fundamental consumer processes to marketing stimuli.

Measurement of the Means-End Chain Model

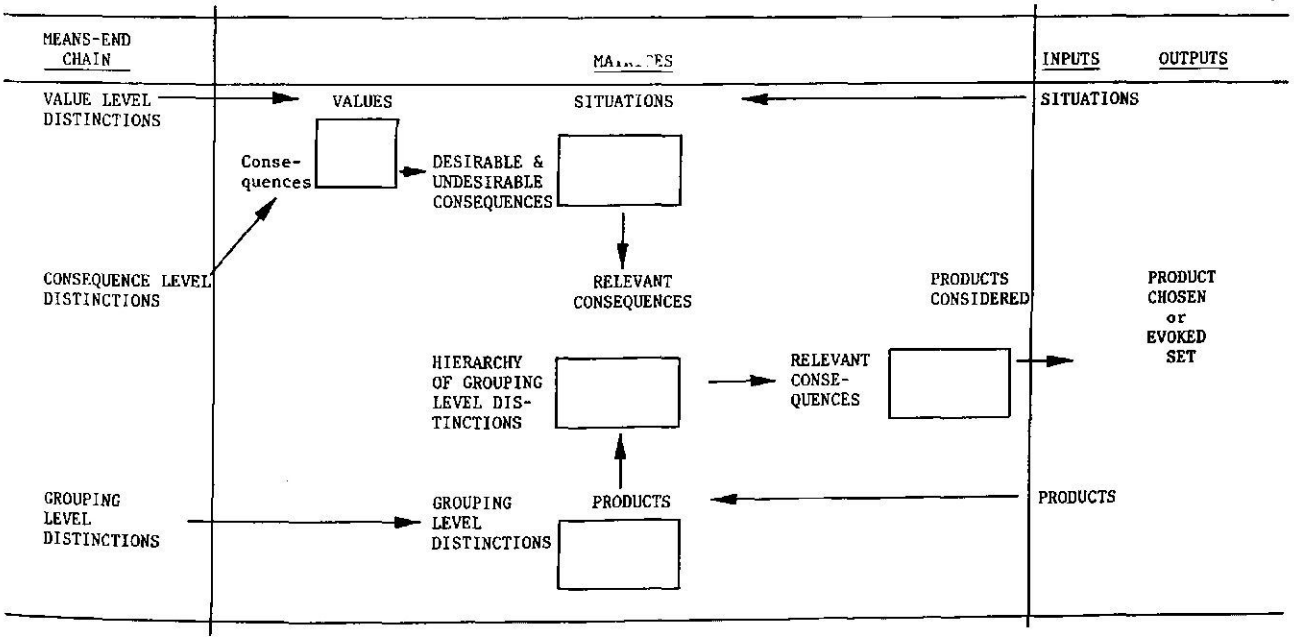
Overview

Figure 3 presents the measurement model for the means-end chain. The figure is divided into three parts: the means-end chain, matrices, and inputs and outputs. The figure shows that the means-end chain is comprised of three levels of distinctions—values, consequences and grouping. The focus of distinctions at the values and consequences levels is on what products can do for the consumer. Distinctions at the grouping level focus on product attributes. Inputs to the system are products and situations; outputs are products chosen for final consideration. The matrices represent the ways in which products and situations are categorized using the distinctions at the various levels in the means-end chain.

Values-Level Distinctions

The values-by-consequences matrix represents the way in which the accumulated experience of the consumer may be represented. The matrix is used to determine which consequences are desirable and which are not. Although values influence behavior, consumers may not be aware that this is taking place. Values, as preferred end-states of existence, are probably not in the forefront of consumer awareness.

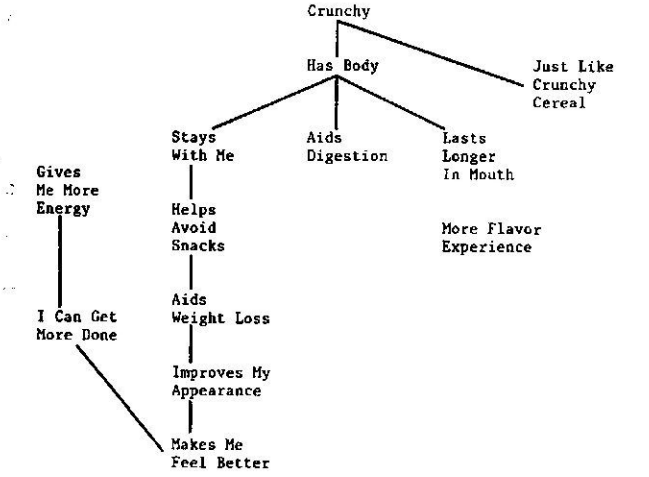
FIGURE 3
Means-End Chain Model



Also, the author is aware that not every consumer act is going to be linked to values. The area of low-involvement marketing (Maloney and Silverman 1979) speaks to this issue. The critical problem with low-involvement products in terms of the model is the lack of linkage between consequences from causing the product and a consumer's values.

To ascertain consumers' values and their relation to consequences and attributes, some method is needed to force consumers to think in value-related terms about their choices. An approach used by Gutman and Reynolds (1979) for deriving values-level distinctions involves the use of the repertory grid (Kelly 1955). Application of the grid requires that respondents state how two products of a set of three are similar and how they differ from the third product (this typically produces attribute-level distinctions). After obtaining several distinctions at this level using this technique, a laddering procedure is used to generate values-level distinctions. Respondents are asked which pole of their dichotomous distinctions they most prefer. Then they are asked why they most prefer the pole they do. The answer to the question typically leads to distinctions involving product functions and consequences from use that become the basis for generating still higher level distinctions. The procedure is repeated until respondents can no longer answer the question. Quite frequently respondents reach a level where they are talking about terminal values such as those indicated by Rokeach. An example of the output from such a procedure is shown in Figure 4, which is based on initially asking people why they prefer crunchy cereal.

FIGURE 4
Connotative Conceptual Structure for Cereals



Situational Influences

Situations encountered by consumers are assessed in terms of consequences that will occur in each. Respondents are asked to indicate which consequences are likely to occur in which situations. This results in a set of relevant consequences emerging that will guide behavior in a given situation. Rarely can all of a consumer's desired consequences be achieved in one situation, and, frequently, achieving some desired consequences means not achieving other desired consequences or having to put up with some undesired consequences. This point of view is consistent with the conclusion that the situational influence is a persuasive factor in consumer behavior (Belk 1975, p. 161).

Grouping-Level Distinctions

The grouping level is the level at which meaning is ascribed to products by grouping them together with other similar products and giving them a group label. Physical and surface properties of products are the bases for grouping products at the grouping level. As the most immediately perceived aspects of products, they serve to identify them in their most fundamental sense. Cars are to drive; refrigerators refrigerate; stoves cook; peas, green beans, carrots are to be eaten. It is possible to discriminate products more finely within each of these categories. There are many ways of classifying cars, many types of refrigerators and many kinds of vegetables.

At the lowest levels of inclusiveness, products are grouped together in accordance with Howard's notion of semantic structure or Lunn's (1972) organization of ways of satisfying consumer needs by product types, product variants and brands. Product types satisfy different needs beyond generic needs (toothpaste vs. toothpowder, hot vs. cold cereal). Product variants are variations within types (with or without fluoride, presweetened vs. not presweetened). And brands represent minor differences within variants. Thus brands as specific alternatives are at the bottom of the hierarchy and are subsumed by product variants, and product variants, in turn, are grouped by product type.

At the upper levels of the hierarchy, the model assumes that evaluative categories subsume product class categories. An evaluative category interposes between consequences that are properties of the person and attributes that are properties of the product. This is a result of the need for the grouping hierarchy to relate to relevant consequences. If "time savings" is sought in a food product, something that is "easy to use" might be sought. "Easy to use" is not a property of a product per se (there is no "easy to use" in a product), and it is not a consequence (something that happens to a person), like "time savings" is a consequence.

Thus in Figure 5, which shows one person's grouping hierarchy for breakfast beverages, soft drinks subsumes colas, lemon-lime and other flavors. At higher levels in the grouping hierarchy, the level of a distinction is determined by its ability to separate products into preferred and nonpreferred sets (see Tversky's theory of sequential choice, 1972). If "not easy to prepare" meant a product was not acceptable for use regardless of any other properties, it would be more functional to have it subsume "good for you"–"not good for you." However, if a consumer would be more likely not to consider a product if it were "not good for you" even if it were easier to prepare than "good for you," "not good for you" would be at a higher level in the hierarchy. The goal of categorizing products at the grouping level is to identify products and array them into sets based on their possession of attributes that imply the ability to provide desired consequences for the consumer.

Two approaches seem most suitable for determining the distinctions that comprise a consumer's grouping hierarchy. A direct approach may be used (Green, Fox and Camone 1969) involving hierarchical merging. The respondent is asked to categorize the products into groups so as to make the products within groups as similar as possible and different from products in other groups. The respondent is asked to merge the two groups he or she considers most similar and to repeat the process until only one group remains. Group labels at each stage provide the distinctions for the grouping hierarchy. (Large groups could also be broken up using the same process in reverse.)

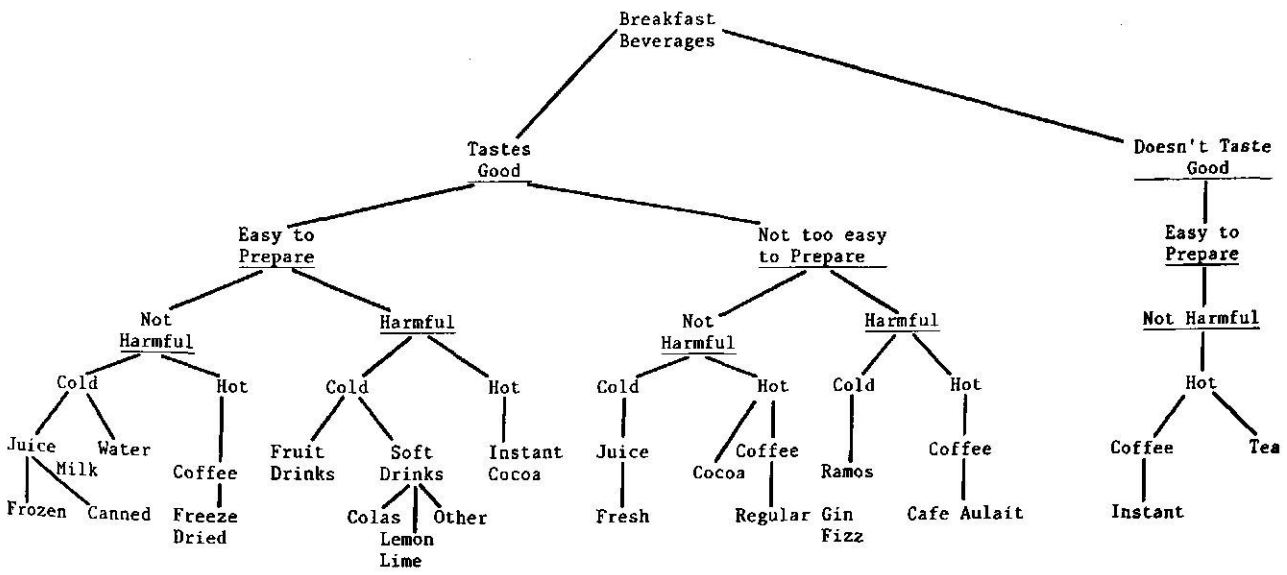
Another approach (Reynolds 1981) begins by eliciting distinctions from respondents using the modified repertory grid procedure and the ordering procedure (both described earlier). The respondent then fills out a distinctions-by-products matrix, where a 1 indicates that a product would move a person toward that pole of a distinction and a 0 indicates that it wouldn't. A procedure described as multidimensional Guttman scaling is used to find Guttman-like chains or relations among distinctions in this matrix. These chains, then, would constitute the distinctions comprising the grouping hierarchy.

Consequence Level Distinctions

As discussed previously, desirable consequences (benefits) derive from values. If a person values "social recognition"—one of Rokeach's values—he/she will develop categories of objects that lead to an increase in social recognition. Benefits such as prestige, sex appeal, white teeth, curly hair and countless other benefits will act as the basis for distinctions that categorize together objects that satisfy these criteria.

One does not literally build benefits into a product. Physical product attributes that provide the benefits (e.g., cost, size, design) are built into a product. It is essential to develop the correspondence between distinctions at the consequence level (e.g., prestige) and the product attributes that imply that the product can provide such benefits. These relations are devel-

FIGURE 5
Grouping Hierarchy for Breakfast Beverages



oped as consumers learn about products from packaging information, advertising, discussions with others, and product trial. The product attributes that imply the ability of the product to provide benefits become key defining attributes for the consumer.

Many relations between attributes and consequences are clear-cut and easy to determine. For beer drinkers, ability to control weight is clearly tied to caloric level in beer, resulting in the category of light beer as opposed to regular beer. In some cases the relationships between consequences and attributes are not as clear or easy to determine. The complexity of or implicative distance between a distinction at the consequence level and product categories at the grouping level require intermediate distinctions between these desirable consequences and the actual product attributes which are so much a part of distinctions at the grouping level.

A consumer who wants good health *and* sees the connection between diet and health may form a natural-artificial distinction. Such a distinction could be applied to drinks by comparing drinks grouped by type (fruit juice, fruit drinks, water, milk, or soft drinks) into an ordered array of categories along the dimension defined by the natural-artificial distinction. An ordered array is a related set of categories whose boundaries are ordinally scaled with respect to the distinction of interest (Green & Tull 1978, p.173).

The sorting of products into categories using distinctions can be achieved by printing one pole of each distinction on a small card and asking respondents to put products into groups that reflect equal amounts of whatever dimension is specified by the distinction (see Bourgeois, Haines and Sommers 1979; Gutman 1977; Gutman and Reynolds 1977, 1979, for examples of this methodology).

Each of the drink-types could be kept in its own category or recategorized according to its degree of artificialness-naturalness, regardless of other properties. The manner in which this would be done is a function of the object properties and the values of the consumer. In some cases such as drinks, the relations between distinctions at the grouping, consequence and value levels may be clear, but there are products where people may have great difficulty judging the alternatives.

Thus, in clothing, for example, where many people are not capable of determining quality of styling without the validation of others, surrogate indicators such as retail outlet are of primary importance. A consumer placing importance on the value of social recognition might seek the benefit of prestige from his/her clothing selections as a means of gaining social recognition. Such a benefit might be implied by clothes judged to have high styling (an evaluation of a property of the clothing), and high styling might be implied by the fact that a certain brand of clothing is carried by the finest stores. There are other bases from which this might be inferred, such as advertised in *Vogue*, worn by the right people, or even intrinsic good taste (in the same sense one learns to identify and appreciate artistic creations). Thus the following distinctions might represent this consumer's means-end chain:

Levels	Distinctions
Value	Social Recognition
Consequence	Prestige
Grouping	High Styling
Grouping	Carried by Finest Stores

The same progression may be examined from the perspective of levels of inclusiveness. Prestige is but one of many ways to get social recognition, wearing high-styled clothing is but one of many ways to obtain prestige, and shopping at the finest stores is but one of many ways to identify high style in clothing. Such behavior has potential negative consequences too. Shopping at the finest stores costs lots of money. The consequences associated with requiring lots of money when you don't have it are, among others, anxiety. And anxiety has its consequences, which are exacted physiologically. These physiological consequences may produce substantial conflict with other desired end-states.

The means-end chain is focused on the linkages between where a person wants to be and the means chosen to get there. A product is indeed a bundle of attributes, and people do consume products for the desirable consequences they get (trading off with any undesirable consequences). It is the implicative relations between product attributes and consequences and between consequences and values that are the objects of study. How these products are grouped at various levels in the means-end chain tells us how each product fits together with other products in the context of the consumer's goal-oriented behavior.

To create a set of products to be considered in any given situation, distinctions at the grouping level are evaluated by the relevant distinctions at the consequence level. The output from this Relevant Consequence Distinction by Grouping Level Distinction matrix is the determination of a path through the grouping level distinctions hierarchy which specifies the set of products to be considered. This set of products is then itself evaluated by the Relevant Consequence Distinctions, resulting in the product or products to be chosen in the situation.

Distinguishing Distinctions

At this point in the model, the consumer has generated a product or set of products for consideration, often referred to as an evoked set, by identifying those

products possessing attributes that imply the ability to provide benefits which in turn are instrumental to the achievement of desired ends. If this process doesn't result in one product being preferred to all others, further processing is still needed. However, similarity has been established, both in terms of the products' physical properties at the grouping level and their ability to provide similar advantages, which lead toward desired values at the benefit level. The basis for this similarity has no further diagnostic use in distinguishing among products at this stage.

Therefore, values cannot further influence choice directly or indirectly. One is left with simple preference (just prefer one to the other) or stochastic choice, wherein variety results by random selection among alternatives or is based on low level choice among insignificant product differences that have no stability over time.

An Illustration of the Model

For measurement purposes, the means-end chain can be conceived of as a series of connected matrices. Using the product category of drinks as an example, the following matrices need to be generated to implement the means-end model described above: a Values-by-Consequences matrix, a Consequence-by-Situations matrix, a Relevant Consequences-by-Grouping Distinctions matrix, and a Relevant Consequences by Product matrix.

This system is illustrated in Figures 6 and 7. Each of the matrices is described below. The approach can be outlined without becoming embroiled in the methodology for generating the data. Only a few distinctions are shown at each level to keep the illustration at a manageable level.

Matrix 6A contains the relations between the consequences deriving from consuming drinks and the values consumers hold. The 1's in the matrix are "signed" to indicate the relation between values and consequences. Some of the situations during which drinks are consumed are shown along the top of Matrix 6B. Consequences to which the consumption of drinks in these situations might ultimately be related are shown on the side of Matrix 6B (along with their

FIGURE 6
Illustration of Means-End Chain

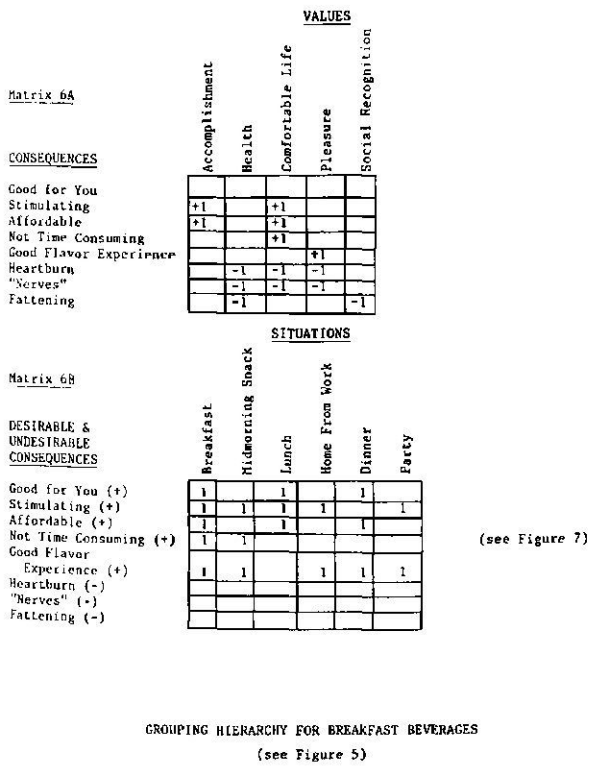


FIGURE 7A
Grouping Level Distinctions by Relevant Benefits Matrix

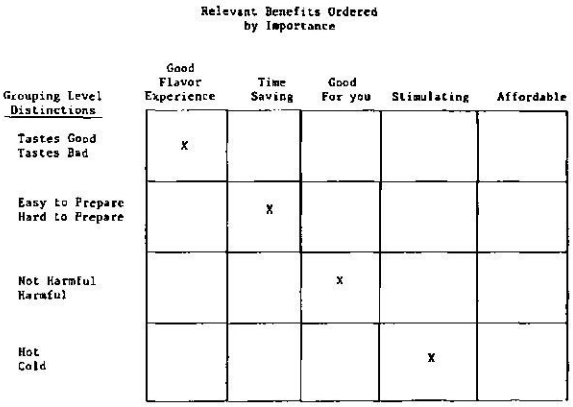
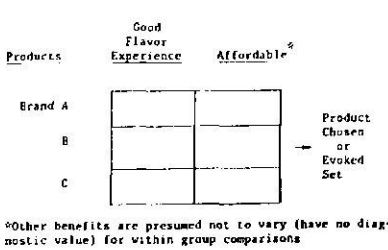


FIGURE 7B
Freeze-Dried Coffee Products Considered by Relevant Benefits



algebraic sign from Matrix 6A). The 1's in the matrix indicate which consequences are relevant in which situations. If breakfast is the situation of interest, the columns of the Situations-by-Consequences matrix for which there were 1's in the "breakfast" row become the columns of the Relevant Consequences-by-Products matrix. The result is that the following consequences are deemed relevant to the choice of a drink in the breakfast situation: good flavor experience, time saving, good for you, stimulating, affordable.

It is unlikely that the values operating in any given situation are immediately and directly apparent. A series of steps are exemplified by the laddering procedure which interposes between one end of the means-end chain and the other. The purpose of these techniques would be to determine the distinctions comprising the chain and the connections between them.

The undesirable consequences (indicated by minus signs) are included in matrix 6B, although they would hardly be sought by consumers in these situations. Their absence, however, may be desired by some consumers who are conscious of products providing these undesirable consequences. In such cases, one could put a "not" in front of the consequence and change the undesired consequence (−) into a desired consequence (+).

The lower part of Figure 6 refers to Figure 5, which showed a hierarchy of products at the grouping level. The distinctions are shown with the products categorized by them. It shows the identity (or product class categories) and evaluative categories.

Figure 7A shows the grouping distinctions by relevant consequence matrix. Both grouping distinctions and relevant consequences are ordered in terms of importance. The determination of the final set of products to be considered is made comparing products categorized by each side of the distinctions in terms of consequences provided deriving from their consumption. Row 1 of Figure 7A shows that products that taste good provide the desired consequence of good flavor experience. Those products that taste good are compared for ease of preparation. Easy-to-prepare products provide the benefit of time saving. Among easy-to-prepare products that taste good, not harmful ones are preferred to harmful ones because they provide more of the benefit good for you. (Some consumers may discriminate among drinks by seeking those that are nutritious or actually good for you; others as above, may have a less stringent criterion; still others may not discriminate among drinks on this basis at all.) Lastly, among not harmful, easy-to-prepare drinks that taste good, hot drinks are preferred at breakfast because they provide the benefit of being stimulating.

Figure 7B shows the relevant consequences made possible by drinks within the tastes good, easy to prepare, not harmful, hot portion of the grouping hierarchy. Comparison of specific alternatives by relevant consequences shows that freeze-dried coffee is the preferred breakfast drink. If cold as well as hot drinks provided relevant benefits, the choice set would have included juice, milk and water in addition to coffee. If more than one product emerges after this screening process, these specific products would be compared for any differences in their attributes to promote the desired benefits. If at the end of this process there was still more than one alternative, some choice process would have to be made, but it wouldn't be made on the basis of the values orientation that controls the structure of the means-end chain.

The selection of one product or the consideration of specific products at this stage may entail some undesired consequences that come along with those products. The consumer would have to trade-off these undesired consequences when considering the desirable consequences.

Applications

The translation of the interest in values to actual research on values has been retarded by the lack of an integrated theory by which values are held to mediate behavior. The model presented in this paper can serve as a catalyst for such research. Building values into marketing planning creates the potential to target products and messages more closely to valued states or goals considered by consumers to be of prime importance.

The means-end chain model offers researchers a guide to procedures that specifically address the linkages connecting values important to the consumer to specific attributes of products. Thus consumers' perceptions and evaluations of products can be studied at different levels in the means-end chain. Consumers tend to operate at the lowest level of abstraction (the grouping level) when sorting products into categories on whatever basis they choose (Gutman 1977). This model and the accompanying measurement procedures represent a way for researchers to systematically link attribute level distinctions used to form product classes to distinctions at the consequence and values level where the basic directions of consumer choice are determined.

Vinson, Scott and Lamont (1977) have outlined several areas where values can contribute to marketing management. It might be useful to point out how the means-end chain model presented herein can contribute in these areas.

Market Analysis and Segmentation

Such a perspective would be valuable in determining product markets from a consumer point of view. Various researchers have defined product markets ranging in breadth from interbrand competition (Butler and Butler 1970, 1971) to competition among widely divergent alternatives (Moran 1973). The ability to examine substitutability at the various levels of the means-end chain can provide a method for such considerations and could aid in identifying competitors that might otherwise be overlooked.

Segmentation research seeks to find groups of consumers defined in terms of demographics, attitude, lifestyle, geographic area or other variables that have a differential response to marketing control variables (Wind 1978). The means-end approach can determine the influence of values for consumer segments and relate these values to distinctions at the consequence and grouping levels.

Specific choices among alternatives are not determined at the values level. But the spectrum of available alternatives can be divided into groups containing objects with the potential to satisfy values and into groups without value-satisfying alternatives. Subgroups of consumers can be established who categorize a product not currently used in value-satisfying categories. Also, a segmentation approach can separate current users of a product class or brand from potential users. The means-end chain approach can identify how consequences and product properties are linked to relevant values by members of these two segments.

Product Planning

The linkage among the levels of the means-end chain ties physical product attributes to values important to consumers. Knowing which product attributes are used by consumers to infer the presence of desired consequences permits clearer specifications for product development by new products departments in their charges to research and development.

Distinctions and their role in forming categories of products can aid managers in determining which competitive brands/products a brand is categorized together with at any level in the means-end chain. The level in the grouping hierarchy at which a product is first grouped together with another product will determine its basic similarity to that product. At the grouping level, products/brands are categorized more on the basis of physical similarity. At the consequence level, categorization is based on functional equivalence in terms of advantages provided to the consumer by consumption of the product. Also, one can continue up to the values level of the means-end chain and group a product with other value-satisfying alternatives. Knowledge of linkages between physical product attributes and consumer benefits and values are essential if marketing managers are to specify clearly desired products' characteristics to research and development.

Promotional Strategy

Establishing linkages between distinctions at all levels in the means-end chain has much potential for advertising. The Grey Benefit Chain was itself developed to aid advertisers in generating effective ad claims. The approach described in this paper extends the scope of the central idea of the Grey Benefit Chain. If an advertiser knew what key distinctions consumers made, how these distinctions were used to establish product/brand groupings, and how this process was carried out at each level in the means-end chain, he/she would be better able to position advertising against the fundamental value orientations of target consumers. (This approach also offers the possibility of increasing the involvement level of low involvement products.)

Research on the Grey Benefit Chain and findings by Gutman and Reynolds (1979) suggest that consumers use surface properties of products as their initial basis for categorization of products. At the same time, if one accepts values as influencing consumer choice, the linkages between the two levels become very important. The author's view of the means-end chain is that advertising research procedures often focus on consumers' low level grouping distinctions without recognizing the hierarchical structure that forms the basis for connecting these distinctions to higher levels in the hierarchy. Similarities and differences are two sides of the coin—we can better understand how products and brands differ when we understand how they are the same. The context created by situation interacts with values to select distinctions at an appropriate level. These processes have to be understood if we are to communicate effectively with consumers.

Conclusions

The means-end model presented in this paper is based on the distinctions consumers use in grouping objects by virtue of their similarity and in distinguishing among them by virtue of how they vary along common dimensions. The model is hierarchical in that it links values that are the ends to basic feature components of products/brands representing the means.

Methodology has only been touched upon, although sorting procedures, repertory grid techniques, and laddering were all mentioned as data-gathering techniques. To a great extent, the model has an ideographic as opposed to a nomethetic orientation. One

has to be able to understand individuals before he/she can understand groups of individuals (Lewin 1951). This, however, doesn't preclude use of scaling or clustering procedures or other types of aggregate analytical techniques. There are some forms of analysis uniquely suited to the binary or categorical data that are typical outputs from the methodologies noted above. Among these, logical construct analysis, akin to multidimensional scaling (Gutman and Reynolds 1977, Reynolds 1981) has shown much promise.

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